

SignFind: A Synchronized Sign Language and Chinese Character Teaching Game for Chinese Deaf Children Using Gesture Recognition

Jintao Nie*
Zhejiang University, China
niejt@zju.edu.cn

Zheng Xu
Zhejiang University, China
22021297@zju.edu.cn

Preben Hansen
Stockholm University, Sweden
preben@dsv.su.se

Ge Yan
Zhejiang University, China
yange@zju.edu.cn

Fangtian Ying
Zhejiang University, China
13067848819@163.com

Yijun Zhao*
Zhejiang University, China
zhaoyj@zju.edu.cn

Jiayan Chen
Zhejiang University, China
chenjiayan@zju.edu.cn

Jianhui Liu
Zhejiang University, China
greyljh@zju.edu.cn

Cheng Yao
Zhejiang University, China
yaoch@zju.edu.cn

Xiubo Liang
Zhejiang University, China
xiubo@zju.edu.cn

Bing Yao
Zhejiang University, China
yaob@zju.edu.cn

Changchao Yu
Zhejiang University, China
yucc2020@zju.edu.cn

Jiadi Wang
Zhejiang University, China
wjd1024@zju.edu.cn

Leijing Zhou
Zhejiang University, China
leijing@zju.edu.cn

Guanyun Wang[†]
Zhejiang University, China
guanyun@zju.edu.cn

ABSTRACT

This paper presents a targeted system to help Chinese deaf children learn both sign language and Chinese characters in early language learning. The system combines sign recognition and in-air writing techniques with games so that children can practice sign language and Chinese character writing at the same time. Sign recognition is used to assess the accuracy of sign language and in-air writing records the process of writing Chinese characters. In addition, the game adds to the learning fun and makes children more willing to complete learning tasks. We have developed a prototype to evaluate the effectiveness of a simultaneous Chinese sign language and Chinese character teaching system based on gesture recognition and in-air writing. We expect that this system will increase children's willingness and efficiency to learn sign language and Chinese characters, and eventually be used to assist in early language education for deaf children in the Chinese region.

*The first two authors contributed equally to this work.

[†]Corresponding authors

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519697>

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools; • **Social and professional topics** → Professional topics; Computing profession; Assistive technologies; User characteristics; People with disabilities.

KEYWORDS

Chinese Sign Language, Chinese Characters, Gesture Recognition, Gesture Interaction, In-air Writing, Gesture-based Gaming, Teaching Methods

ACM Reference Format:

Jintao Nie, Yijun Zhao, Bing Yao, Zheng Xu, Jiayan Chen, Changchao Yu, Preben Hansen, Jianhui Liu, Jiadi Wang, Ge Yan, Cheng Yao, Leijing Zhou, Fangtian Ying, Xiubo Liang, and Guanyun Wang. 2022. SignFind: A Synchronized Sign Language and Chinese Character Teaching Game for Chinese Deaf Children Using Gesture Recognition. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519697>

1 INTRODUCTION

For children with deafness, it is critical to expose children to sufficient language examples to help develop lifelong language skills during this critical period of language development [24]. Data show that over 95% of parents of deaf children have normal hearing and they are unable to teach sign language, resulting in a lack of adequate sign language learning opportunities for deaf children in the home environment [2, 12]. Early education for children is very important [11]. Due to the current underdevelopment of China's

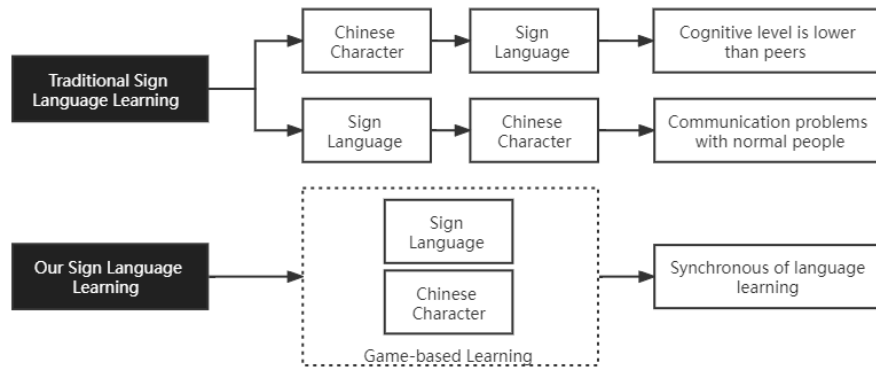


Figure 1: Comparison of sign language teaching mode

special education system and the limitations of parents' perceptions of deaf children, the hearing impaired children tend to receive Chinese language education first in the language learning process, and sign language usually becomes the second language they learn. This lack of synchronization has caused deaf children to miss the critical period of language learning and to lag behind hearing children in their perception of the world. In recent years, the field of special education in China has become aware of the issues raised by sign language as a second language for deaf children and has gradually started teaching sign language as the first language for deaf children and Chinese as the second language [25]. However, there are problems in the existing teaching model. For example, in the context of Chinese language, deaf children who are taught sign language first and ignore the learning of Chinese may have difficulties in communicating with hearing people. By learning sign language as their first language, they may also tend to be satisfied with the only knowledge of sign language for communication in the closed environment of special education schools and lose the motivation to learn Chinese further [8]. In addition, Chinese is considered one of the most challenging human languages in the world, and Chinese is often considered a morphological language, not an alphabetic language [9, 22]. Furthermore, Chinese characters are the smallest independent writing units in the Chinese language. Each character has a square stroke pattern, which is very different from the linear structure found in other alphabetic languages [10]. Finally, learners of Chinese as a second language need to remember not only the gestural expressions of the characters, but also their unique writing structures. The complexity of Chinese characters makes learning Chinese as a second language a daunting task for learners [23]. In Chinese primary education, teachers teach students to write Chinese characters by using their fingers instead of pens, moving their fingers in the air as if they were writing, in order to deepen their memory of the shapes of Chinese characters [20, 21].

In this paper, we propose an educational video game to help Chinese deaf children learn sign language and Chinese characters simultaneously. Figure 1 illustrates the differences between the traditional sign language teaching model and our proposed teaching model. The learning system uses gesture recognition [2, 6, 17]

and in-air writing technology. Children will enter the game scenario when using the game, interact with the game elements by controlling the game character, trigger the opening of the camera to practice Chinese sign language and write Chinese characters, and collect all the game elements to reach the destination to succeed in the level. In order to explore the possibilities of SignFind in language education for hearing impaired children in China, and to verify whether it can help them improve their willingness to learn and enhance their learning outcomes in sign language and Chinese character learning. We completed the design of the system and tested it on four deaf children. The results show that encouraging deaf children to learn and practice Chinese sign language and Chinese character writing at the same time through the game helps them to master Chinese sign language and Chinese character writing faster and better.

2 RELATED WORK

There has been some research on the application of games to sign language teaching systems, and it is considered to be a way to motivate and engage students to increase their engagement and motivation when learning to interact. Lee et al. designed a gesture recognition-based American Sign Language (ASL) game in which children need to make the correct gestures on cue to interact with virtual characters [7]. Parreño et al. designed a crossword and tic-tac-toe based game for teaching Ecuadorian sign language by detecting gesture signals in Unity3D software and interacting with the user in a virtual environment [17]. Madeo designed a game to recognize Brazilian language, extracting frames from videos containing gestures and then processing them through Matlab R. If the gesture is wrong the game character is punished [14]. Jiajia Li et al. designed a game of AR combined with gesture recognition [10]. Amma et al. proposed a wearable input system that enables users to write text in the air through 3D handwriting recognition [1]. In addition, several research projects have evaluated different sign language input tools and recognition models. Mohandes et al. used Leap Motion to recognize Arabic sign language and compared the performance of a multilayer perceptron (MLP) neural network with a Nave-Bayes classifier [13]. Bach et al. proposed a framework for Vietnamese language recognition using recurrent neural networks

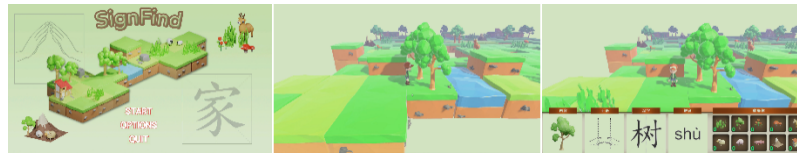


Figure 2: Game interface

(RNN) and Mediapipe gesture tracking [4]. Brashear et al. trained Hidden Markov Models for gesture recognition by recognizing information from colored gloves and wireless accelerometers [3].

In our system, the sign language and Chinese characters that children need to practice are determined by the virtual game elements in the game. When a child encounters a game element, he needs to make the correct sign language gesture or write the correct Chinese character in the air to collect it in order to move on. In terms of technology, we used Mediapipe hand recognition framework, which not only captures and recognizes hand gestures quickly through the camera, but also supports overhead writing. Based on these two technologies, we designed a contextual adventure game in which users need to control virtual game characters to find their way home in the forest and practice sign language postures and Chinese character writing in the process. The game is a novel attempt to help Chinese deaf children to be better bilingual at the early stage of language learning.

3 THE SYSTEM

SignFind is an educational computer game based on the real-time hand tracking technology provided by Google Mediapipe. The game runs on a screen and camera-equipped computing device and is used to develop Chinese Sign Language (CSL) and Chinese character writing skills for children ages 5-11. SignFind's goal is to encourage Chinese deaf children to practice both sign language postures and Chinese character writing in a fun and engaging way, thereby increasing their willingness and effectiveness to learn. The game is interactive. It requires the player to interact with the computing device by means of keyboard operations as well as gestures. And it includes game elements that match the correct sign cues, real-time camera capture (providing data input to the sign recognition and in-air writing system and real-time feedback to the child through the interface), animated characters that can be controlled by keyboard movement, and game obstacles that can receive feedback from the child's sign and writing commands. The focus of this game is to practice and correctly repeat Chinese sign language and Chinese character writing, allowing the child to communicate with the system through Chinese sign language and aerial Chinese character writing.

3.1 Game Design

The design of the game interface conforms to the Nielsen Ten Principles of Interaction [15, 16], as shown in Figure 2. The research questions include game content, level, character, and gameplay design. Players need to control the game character to move in the map to reach the final destination, and in addition, obstacles are set as game elements to obstruct the player's progress. These obstacle elements are mapped with real-life objects, and they are cleverly

designed as sign language and Chinese characters that players need to learn in the course of a single game session. Once the game starts, players can use the computer keyboard, mouse and gestures to operate the game. The flow chart of the game is shown in Figure 3

3.2 Sign Language and Chinese Character Data

With the guidance of a teacher who teaches hearing-impaired children, and after reviewing the Chinese sign language and Chinese language textbooks of Chinese special education schools at the early age, we selected some sign language and corresponding Chinese characters that need to be mastered at the early age, and created virtual obstacles mapped to real life, as shown in Figure 4

3.3 Bilateral Communication

In the system SignFind, Unity implements the game-related development and python implements the gesture recognition and in-air writing algorithms with real-time hand tracking with the camera on. Unity and Python need to communicate bilaterally as shown in Figure 5, i.e., the two developed environments read and write data simultaneously. Unity needs to transmit the player's commands to Python during the game so that Python can turn on the camera at the right time and recognize the key points of the player's hand features for tracking, gesture recognition or in-air writing. After the input is finished, the data will be matched with the parameters transferred from Unity, and the data will be returned to Unity whether the matching result is correct or not, to realize the elimination and retention of game elements.

3.4 Data Processing

After the camera of the computing device is turned on, the data processing process is regarded as the coordinate position of the key point of the input user's finger, and the data is processed in real time for interaction with the game scene. Using a camera to track hands and fingers, Google provides the Mediapipe Hands and Fingers Tracking API. The Mediapipe used machine learning (ML) to infer 21 3D landmarks of hands from a single frame of images taken from the camera (Figure 6). This technology has certain constraints on the user's hand movements. For example, in the gesture recognition window, the user's hand cannot move too fast or too close or too far from the camera, and the recognition can be incorrect when the hands overlap. It is necessary to ensure that the camera can fully recognize the hand joints in order for the system to function properly.

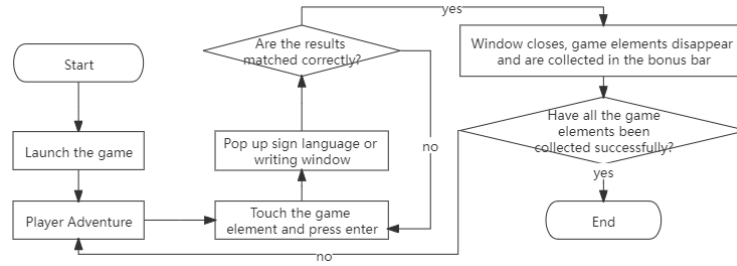


Figure 3: Game flow chart

Game elements	Sign language	Chinese characters	Game elements	Sign language	Chinese characters
grass		草	tree		树
flower		花	pig		猪
cow		牛	mountain		山
mushroom		菇	horse		马
fish		鱼	home		家

Figure 4: Sign language and Chinese characters used in the game

3.5 Gesture Recognition and In-air Writing

In gesture recognition algorithm, different detection algorithms are set for different gesture features [2]. For example, the mountain gesture in Figure 4 is characterized by: (1) palm outward; (2) thumb, index finger and little finger open, index finger and ring finger closed; (3) one hand; The detection algorithm for mountain gesture will detect whether the gesture matches the above three characteristics, if so, the gesture will be judged as a mountain. The direction of the palm can be judged by the position of the thumb relative to the palm. The degree of finger bending can be judged by the included Angle of finger joint vector. The number of hands is provided by Mediapipe Hands API. The process of gesture recognition is shown in Figure 7

The process of writing in the air is implemented as follows, with the writing mode set to thumb and index finger together and the pen holding mode set to thumb and index finger apart. In this way, the thumb and index finger are put together as a brush and the movement trajectory of the brush is recorded. The movement trajectory is drawn on the picture by OpenCV and displayed to the user in real time. We refer to previous work on Chinese character recognition [18, 19]. Based on our knowledge of related technology applications, we chose Tencent Cloud's Optical Character Recognition (OCR) technology to implement the Chinese character recognition function in our system. It is based on the world-leading deep learning

technology of Tencent Youtu Lab, which can intelligently recognize the text on images as editable text. After the user's handwriting is completed, the technology's AI API is used to recognize the user's handwritten Chinese characters and compare them with the target Chinese characters. For example, if the kanji for flower is written in the air, the result will be recognized as flower (Figure 8) and the result will be returned to Unity to be matched with the game elements.

4 USER STUDY

We engaged 4 children with congenital hearing impairment from special education schools, 2 girls and 2 boys between the age of 5 and 7 in the evaluation of the game (Figure 9). All the sessions were guided by a sign language teacher and an experimenter as tester.

Firstly, we informed the children, their teachers and parents about this study and obtained their consent. Afterward, the children and teachers were asked whether the children had previously acquired the sign language and Chinese characters involved in the game. We were told that they had learned the sign language for "grass", "tree", and "home" in the game, but could not write the corresponding Chinese characters. For "mushroom" and "pig" in the game, they had not learned the corresponding sign language and Chinese characters. They had learned the Chinese characters and sign language for the other five game elements that matched.

We tested the usability of the game with all 4 users. Before starting the test, the sign language teacher informs the child about the basic operation of the game, the distance that the camera could recognize, and the icons representing the sign language or writing before starting the test. During the test, the teacher accompanied the whole process to simulate the general usage scenarios of the game. On average, each child spent 40 minutes playing the game. After playing the game each child filled in a usability evaluation survey comprising of 10 questions related to the quality of the game's interactive mode, game story and difficulty level, using a 5-point Likert scale. Furthermore, we asked children and teachers to describe how they felt during the test.

5 RESULTS AND DISCUSSION

Our results clearly show that the SignFind educational game can improve the willingness and effectiveness of sign language and Chinese character writing among hearing impaired children in China. We made a questionnaire and asked the sign language teachers to relay the questions to the children through sign language. In the

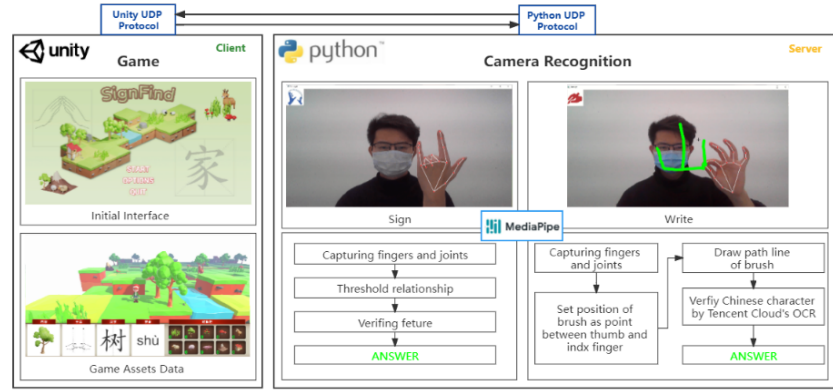


Figure 5: Bilateral communication

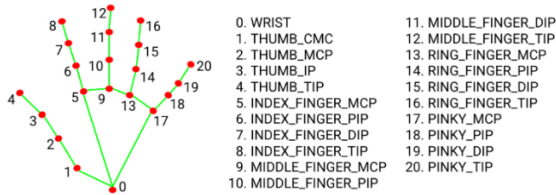


Figure 6: Hand key point landmark



Figure 7: Gesture recognition

children's questionnaire (Table 1), the average score was 4.5 for "how easy it is to understand the rules of the game?", 5 for "Is this game format interesting?", 5 for "Is the game easy to manipulate?", 4.75 for "Will you not feel frustrated when you fail to collect game materials?", 5 for "Did the tips provided by the game help you?". These results indicate that the game format is highly appealing to children and easy to understand and manipulate. They were happy to explore the areas of the game and were very happy to collect game elements by doing the correct sign language and writing the correct Chinese characters. They did not feel frustrated even when the computer misidentified the gestures or misidentified the writing during the process. On the contrary, they were more interested in trying to correct the gesture errors or writing errors after reviewing the hints. The average score was 5 for "Do you not feel tired of learning 10 sign languages and Chinese characters at a time?", 5 for "Would you like to do more exercises using this educational game?", 5 for "Is this game better to help you learn sign language?", 4.75 for "Does in-air writing help to memorize the structure of

Chinese characters?", 5 for "Is it easier to learn both sign language and Chinese characters with this system than without it?". The sign language recognition and in-air writing in the game had a positive impact on children's learning of sign language and memorization of the Chinese character structure, and children felt that they learned better with the system than before. In addition, we also interviewed the sign language teacher. She saw that the game gave timely feedback after the child did sign language or wrote characters. When the feedback result was correct, the child had a smile on her face then kept exploring other areas on the map; while when the feedback result was wrong, the child would go to check the hints and practice trying again and again, which was the first time she saw the child concentrate on correcting her incorrect sign language movements and Chinese character writing repeatedly. These findings show that SignFind has a positive effect on deaf children's learning and practice of sign language and Chinese character writing, increasing their willingness to learn and their learning efficiency.

6 CONCLUSION AND FUTURE WORK

This paper presents a game designed to assist language learning for hearing-impaired children in China. It was developed through unity game development combined with gesture recognition technology and aerial writing technology, and its feasibility was empirically demonstrated through preliminary user studies. However, there are limitations in this study; although many factors were considered in the system design, only a preliminary study was conducted and the performance of each element was not verified. In future research work, we will draw on the experimental approach [5] used in the study conducted by Danielle et al. to recruit more children with hearing loss between the ages of 5 and 11 as an experimental group for evaluation testing against a control group, and we plan to use more quantifiable methods to assess the rate of improvement of the system on children's learning outcomes. The accuracy of the gesture recognition and Chinese character recognition techniques used in the system in this paper also needs to be evaluated and improved to further assist in correcting sign language movements and standardized Chinese character writing. Finally, we will also work closely with deaf and sign language linguists to investigate

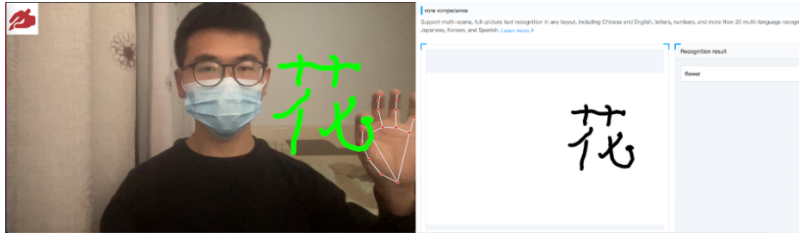


Figure 8: In-air writing process and Chinese character recognition results

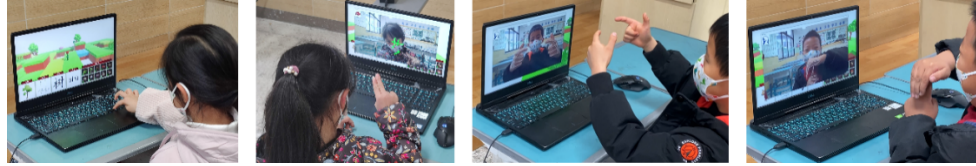


Figure 9: User Test

Table 1: Results of the questionnaire for children

Questions	BoyA	BoyB	GirlC	GirlD	Avg
I. Do you think the rules of the game are easy to understand? (1: hard – 5: easy)	4	5	4	5	4.5
II. Do you think the form of the game is interesting? (1: boring – 5: interesting)	5	5	5	5	5
III. Is it easy for you to manipulate during the game? (1: hard – 5: easy)	5	5	5	5	5
IV. Do you not feel depressed after losing in a game? (1: depressed – 5: positive)	5	4	5	5	4.75
V. Do you think the hints during the game are helpful? (1: useless – 5: helpful)	5	5	5	5	5
VI. Do you not feel tired of learning 10 sign languages and Chinese characters at a time? (1: tired – 5: energetic)	5	5	5	5	5
VII. Would you like to use this system for more practice? (1: no – 5: yes)	5	5	5	5	5
VIII. Does this game help you learn sign language better? (1: useless – 5: helpful)	5	5	5	5	5
IX. Does in-air writing help you remember the structure of Chinese characters? (1: useless – 5: helpful)	4	5	5	5	4.75
X. Do you think it is easier to learn two languages at the same time with this system than without? (1: harder – 5: easier)	5	5	5	5	5

other factors besides gestures that can help children with hearing loss to improve their learning outcomes and seek to improve our system.

ACKNOWLEDGMENTS

This research has benefited from the help and support of many parties. We express our gratitude to Professor Xuan Zheng, from the National Center for Sign Language and Braille Research, School of Special Education, Faculty of Education, Beijing Normal University. Professor Zheng provided professional support in teaching sign language to deaf students in special education schools. We also express our gratitude to Huaxia Publishing House, which provided us with Chinese Universal Sign Language image materials in this study.

REFERENCES

- [1] Christoph Amma, Marcus Georgi, and Tanja Schultz. 2014. Airwriting: a wearable handwriting recognition system. *Personal Ubiquitous Comput.* 18, 1 (January 2014), 191–203. DOI:https://doi.org/10.1007/s00779-013-0637-3
- [2] Dhruva Bansal, Perna Ravi, Matthew So, Pranay Agrawal, Ishan Chadha, Ganesh Murugappan, and Colby Duke. 2021. CopyCat: Using Sign Language Recognition to Help Deaf Children Acquire Language Skills. *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, Article 481, 1–10. DOI:https://doi.org/10.1145/3411763.3451523
- [3] Helene Brashear, Valerie Henderson, Kwang-Hyun Park, Harley Hamilton, Seungyeon Lee, and Thad Starner. 2006. American sign language recognition in game development for deaf children. In *Proceedings of the 8th international ACM SIGACCESS conference on Computers and accessibility (Assets '06)*. Association for Computing Machinery, New York, NY, USA, 79–86. DOI:https://doi.org/10.1145/1168987.1169002
- [4] Bach Duy Khuat, Duong Thai Phung, Ha Thi Thu Pham, Anh Ngoc Bui, and Son Tung Ngo. 2021. Vietnamese sign language detection using Mediapipe. In *2021 10th International Conference on Software and Computer Applications (ICSCA 2021)*. Association for Computing Machinery, New York, NY, USA, 162–165. DOI:https://doi.org/10.1145/3457784.3457810
- [5] Danielle Bragg, Raja Kushalnagar, and Richard Ladner. 2018. Designing an Animated Character System for American Sign Language. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '18)*. Association for Computing Machinery, New York, NY, USA, 282–294. DOI:https://doi.org/10.1145/3234695.3236338
- [6] Eyal Krupka, Kfir Karmon, Noam Bloom, Daniel Freedman, Ilya Gurvich, Aviv Hurvitz, Ido Leichter, Yoni Smolin, Yuval Tzairi, Alon Vinnikov, and Aharon Bar-Hillel. 2017. Toward Realistic Hands Gesture Interface: Keeping it Simple for Developers and Machines. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, Article 100, 1–10. DOI:https://doi.org/10.1145/3025754.3025854

- Factors in Computing Systems (CHI '17). Association for Computing Machinery, New York, NY, USA, 1887–1898. DOI:<https://doi.org/10.1145/3025453.3025508>
- [7] Seungyon Lee, Valerie Henderson, Harley Hamilton, Thad Starnier, Helene Brashar, and Steven Hamilton. 2005. A gesture-based american sign language game for deaf children. In CHI '05 Extended Abstracts on Human Factors in Computing Systems (CHI EA '05). Association for Computing Machinery, New York, NY, USA, 1589–1592. DOI:<https://doi.org/10.1145/1056808.1056973>
- [8] Chen Lv. 2021. Exploring the value of Chinese language teaching in special education. *Chinese Character Culture* (20), 104–105. DOI:<https://doi.org/10.14014/j.cnki.cn11-2597/g2.2021.20.044>
- [9] Lu, Ming-Tsan Pierre, Gregory L. Hallman Jr, and John B. Black. 2013. Chinese character learning: Using embodied animations in initial stages. *Journal of Technology and Chinese language teaching* 4.2 (2013): 1–24. DOI: <http://www.tclt.us/journal/2013v4n2/luhallmanblack.pdf>
- [10] Jiajia Li, Zixia Zheng, Xie-min Wei, and Guanyun Wang. 2021. FaceMe: An Augmented Reality Social Agent Game for Facilitating Children's Learning about Emotional Expressions. The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology. Association for Computing Machinery, New York, NY, USA, 17–19. DOI:<https://doi.org/10.1145/3474349.3480216>
- [11] Shuai Li, Jing Chen, Mengda Li, Jie Lin, and Guanyun Wang. 2017. ColorOdor: Odor Broadens The Color Identification of The Blind. In Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '17). Association for Computing Machinery, New York, NY, USA, 2746–2751. DOI:<https://doi.org/10.1145/3027063.3053186>
- [12] Goldin-Meadow, S. , Mylander, C. , & Franklin, A. . 2007. How children make language out of gesture: morphological structure in gesture systems developed by american and chinese deaf children. *Cogn Psychol*, 55(2), 87–135. DOI:<https://doi.org/10.1016/j.cogpsych.2006.08.001>
- [13] M. Mohandes, S. Aliyu and M. Deriche. 2014. Arabic sign language recognition using the leap motion controller, 2014 IEEE 23rd International Symposium on Industrial Electronics (ISIE). 960–965, DOI:<https://doi.org/10.1109/ISIE.2014.6864742>
- [14] Renata Cristina Barros Madeo. 2011. Brazilian sign language multimedia hangman game: a prototype of an educational and inclusive application. In The proceedings of the 13th international ACM SIGACCESS conference on Computers and accessibility (ASSETS '11). Association for Computing Machinery, New York, NY, USA, 311–312. DOI:<https://doi.org/10.1145/2049536.2049623>
- [15] Jakob Nielsen. 1994. Enhancing the explanatory power of usability heuristics. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '94). Association for Computing Machinery, New York, NY, USA, 152–158. DOI:<https://doi.org/10.1145/191666.191729>
- [16] Jakob Nielsen. 1995. Usability inspection methods. In Conference Companion on Human Factors in Computing Systems (CHI '95). Association for Computing Machinery, New York, NY, USA, 377–378. DOI:<https://doi.org/10.1145/223355.223730>
- [17] Mateo A. Parreño, Carmen J. Celi, Washington X. Quevedo, David Rivas, and Víctor H. Andaluz. 2017. Teaching-Learning of Basic Language of Signs through Didactic Games. In Proceedings of the 2017 9th International Conference on Education Technology and Computers (ICETC 2017). Association for Computing Machinery, New York, NY, USA, 46–51. DOI:<https://doi.org/10.1145/3175536.3175584>
- [18] Ning Xu, Weiqiang Wang, and Xiwen Qu. 2015. A discriminative classifier for in-air handwritten Chinese characters recognition. In Proceedings of the 7th International Conference on Internet Multimedia Computing and Service (ICIMCS '15). Association for Computing Machinery, New York, NY, USA, Article 16, 1–4. DOI:<https://doi.org/10.1145/2808492.2808508>
- [19] Songhua Xie, Hailiang Yang, and Hui Nie. 2020. Design of Chinese Character Recognition Based on AlexNet Convolution Neural Network. In Proceedings of the 2020 3rd International Conference on Artificial Intelligence and Pattern Recognition (AIPR 2020). Association for Computing Machinery, New York, NY, USA, 68–73. DOI:<https://doi.org/10.1145/3430199.3430230>
- [20] Itaguchi Y, Yamada C, Fukuzawa K. 2015. Writing in the Air: Contributions of Finger Movement to Cognitive Processing. *PLoS ONE* 10(6): e0128419. DOI:<https://doi.org/10.1371/journal.pone.0128419>
- [21] Itaguchi Y, Yamada C, Yoshihara M, Fukuzawa K. 2017. Writing in the air: A visualization tool for written languages. *PLoS ONE* 12(6): e0178735. DOI:<https://doi.org/10.1371/journal.pone.0178735>
- [22] Yifan Yang, Leijing Zhou, Rujian Li, Hang Yao, Jialu Song, and Fangtian Ying. 2019. Chinese Character Learning System. In Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems (CHI EA '19). Association for Computing Machinery, New York, NY, USA, Paper LBW2218, 1–5. DOI:<https://doi.org/10.1145/3290607.3312813>
- [23] Zhan, Hong, and Hsiu-Jen Cheng. 2014. The role of technology in teaching and learning Chinese characters. *International Journal of Technology in Teaching and Learning*, 10.2:147. DOI:<https://commons.erau.edu/publication/1097>
- [24] Zahoor Zafrulla, Helene Brashar, Thad Starnier, Harley Hamilton, and Peter Presti. 2011. American sign language recognition with the kinect. In Proceedings of the 13th international conference on multimodal interfaces (ICMI '11). Association for Computing Machinery, New York, NY, USA, 279–286. DOI:<https://doi.org/10.1145/2070481.2070532>
- [25] Zhai Zhou, Yanke Wei. 2021. An analysis of the real-life dilemma of sign language teaching in deaf schools in China. *Journal of Suihua College*. DOI: <https://kns.cnki.net/kcms/detail/detail.aspx?FileName=SHSZ202107013&DbName=CJFQ2021>